MANGALORE REFINERY AND PETROCHEMICALS LIMITED (MRPL)

STATIONARY EQUIPMENT INTEGRITY REPORT - CDU PRE-HEAT TRAIN

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Equipment ID: MRPL-HEX-105-AB

Equipment Description: Crude vs Atmospheric Residue Pre-Heat Exchanger

Operating Unit: CDU-1 Pre-Heat Train (Train B)

Inspection Date: 2026-09-04

Inspector: M. V. Chacko (Senior Static Equipment Inspector, API 510/653)

OBSERVATIONS & THERMAL HYDRAULIC PERFORMANCE:

- Shell-Side Differential Pressure (Residue Stream): 2.45 bar (Clean Baseline: 0.85 bar).
- Tube-Side Differential Pressure (Crude Feed): 1.95 bar (Clean Baseline: 0.90 bar).
- Heat Transfer Coefficient (U-Value): Dropped from design 420 W/m2-K to 215 W/m2-K (48% drop).
- Logarithmic Mean Temperature Difference (LMTD): Severely pinched at 16.2 deg C.
- Ultrasonic Thickness Gauging (UT): Shell wall thickness 14.2 mm (Min retirement: 11.5 mm).
- **Gasket Sealing Zone: Channel head flange weeping observed at south nozzle connection.**

SEVERITY & ROOT CAUSE ASSESSMENT:

Severe asphaltene precipitation and heavy coking in tube bundle passes due to desalter temperature excursion two weeks prior. Excessive hydraulic resistance threatens crude throughput.

RECOMMENDED CORRECTIVE ACTION PLAN:

1. Isolate Exchanger Train B using block valves and divert through bypass loop.
2. Perform on-line high-pressure hydro-jetting (1200 bar water lancing) of tube bundle.
3. Conduct chemical solvent circulation (aromatic gas oil wash) to dissolve asphaltene matrix.
4. Replace Channel Head Kammprofile metallic gasket before post-cleaning hydrotest.